
Technical University of Crete
School of Electrical and Computer Engineering
Telecommunication Systems I
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Exercise 1
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Hours required: 4

PART A

A.1 Construct SRRC pulses $\phi(t)$. Indicative values: $T = 10^{-3}$ sec, $\text{over} = 10$, $T_s = \frac{T}{\text{over}}$, $A = 4$, and roll-off factor $a = 0, 0.5, 1$ (here A denotes half the length of the truncated pulse measured in symbol periods).

(5) Plot the pulses on the same figure (a **common** plot) with an appropriate time axis, for $T = 10^{-2}$ sec, $\text{over} = 10$, $A = 4$, and $a = 0, 0.5, 1$.

(5) What do you observe about the rate at which the pulse amplitude “decays” as the absolute time $|t|$ increases, with respect to the value of a ? Which pulse decays faster?

Matlab code:

```
% Parameters
```

```
T = 10^-2;  
over = 10;  
A = 4;
```

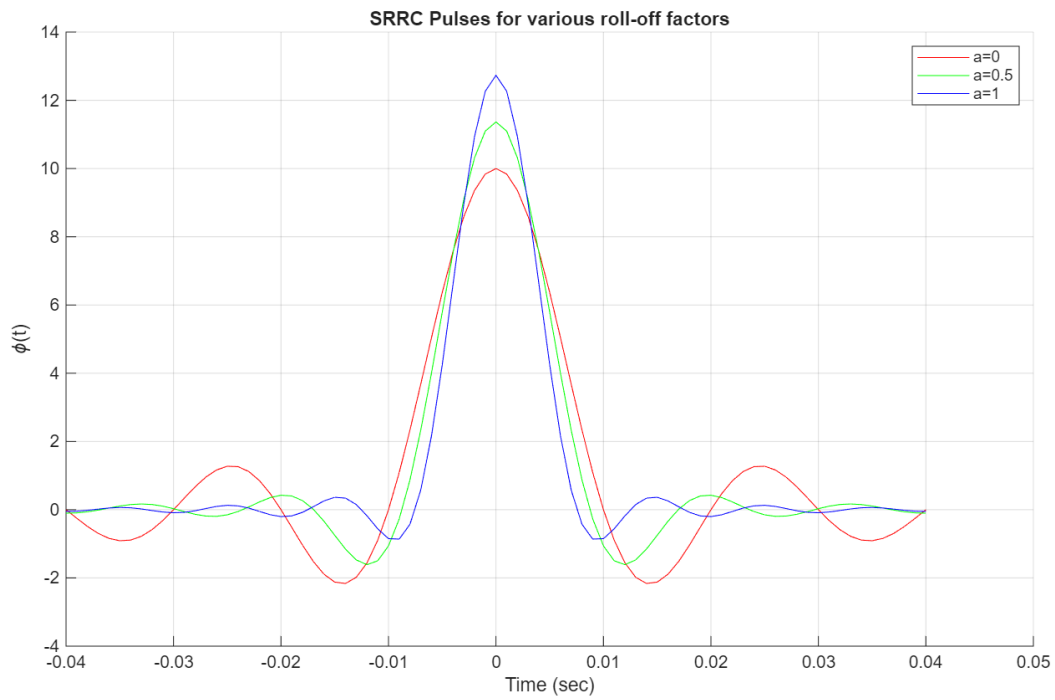
```
figure; hold on;
```

```
[phi, t] = srrc_pulse(T, over, A, 0);  
plot(t, phi, 'r');
```

```
[phi, t] = srrc_pulse(T, over, A, 0.5);  
plot(t, phi, 'g');
```

```
[phi, t] = srrc_pulse(T, over, A, 1);  
plot(t, phi, 'b');
```

```
grid on;  
legend('a=0', 'a=0.5', 'a=1');  
xlabel('Time (sec)'); ylabel('\phi(t)');  
title('SRRC Pulses for various roll-off factors');
```



As the roll-off factor a increases, the amplitude of the pulse "decays" towards zero as the absolute time $|t|$ increases. The pulse with $a = 1$ decays the fastest, while the pulse $a = 0$ has the slowest decay and the largest side-lobes.

A.2 Using the functions `fft` and `fftshift`, compute the Fourier transforms $\Phi(F)$ of the pulses from the previous step at N_f equally spaced frequency points over $[-\frac{F_s}{2}, \frac{F_s}{2})$ (e.g., $N_f = 1024, 2048$).

Plot the energy spectral density $|\Phi(F)|^2$ of these pulses

- (a) (5) on a common `plot`,
- (b) (5) on a common `semilogy` (note that `semilogy` lets you study the values of $|\Phi(F)|^2$ over ranges where they are very small, which cannot be done easily with a linear `plot`).

MATLAB code:

```
% Parameters
T = 10^-2;
over = 10;
A = 4;
alphas = [0, 0.5, 1];
Nf = 2048;
Ts = T / over;
Fs = 1 / Ts;
freq = linspace(-Fs/2, Fs/2 - Fs/Nf, Nf);

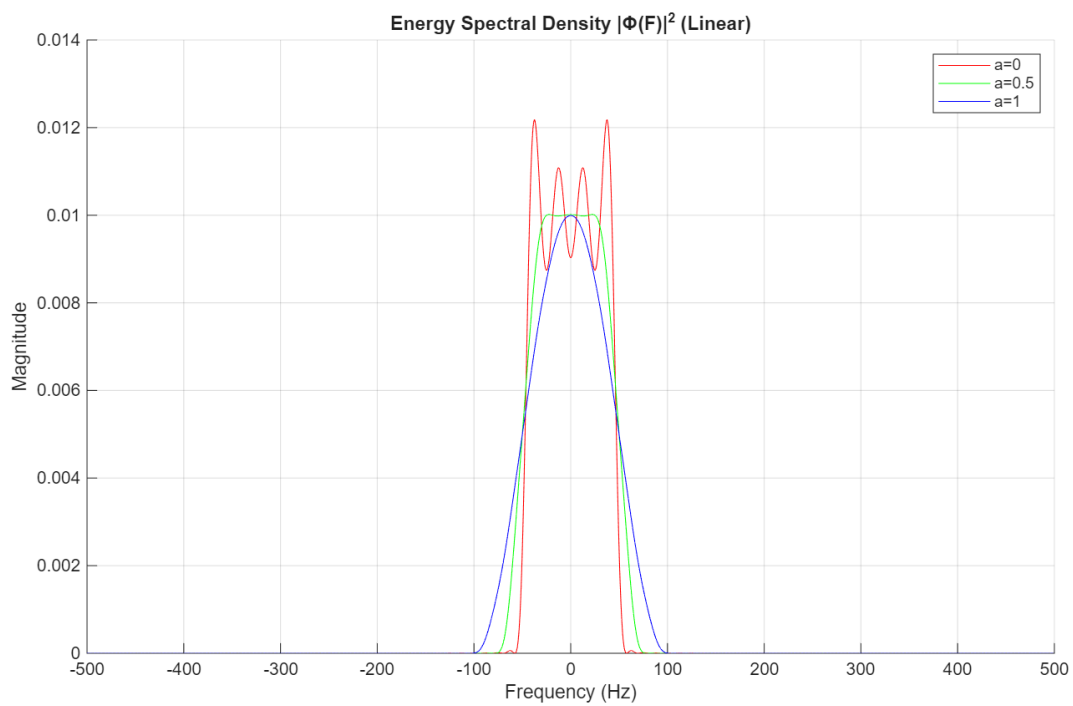
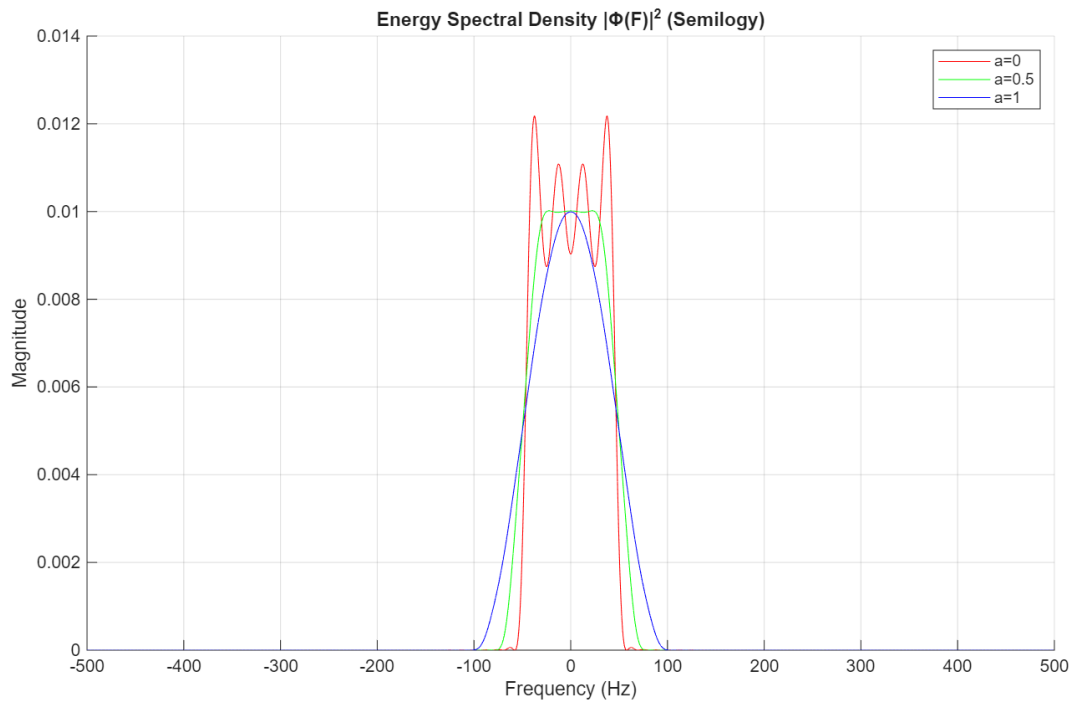
% Initialize figures
fig1 = figure(); hold on; grid on;
fig2 = figure(); hold on; grid on;
colors = ['r', 'g', 'b'];

for i = 1:length(alphas)
    [phi, t] = srcc_pulse(T, over, A, alphas(i));
    Phi_F = fftshift(fft(phi, Nf)) * Ts;
    ESD = abs(Phi_F).^2;          %|Phi(F)|^2

    % (a) Plot on a common linear plot
    figure(fig1);
    plot(freq, ESD, colors(i));

    % (b) Plot on a common semilogy plot
    figure(fig2);
    semilogy(freq, ESD, colors(i));
end

figure(fig1);
title('Energy Spectral Density |\Phi(F)|^2 (Linear)');
xlabel('Frequency (Hz)'); ylabel('Magnitude');
legend('a=0', 'a=0.5', 'a=1');
figure(fig2);
title('Energy Spectral Density |\Phi(F)|^2 (Semilogy)');
xlabel('Frequency (Hz)'); ylabel('Magnitude');
legend('a=0', 'a=0.5', 'a=1');
```



A.3 The theoretical bandwidth of infinite-duration pulses is $BW = \frac{1+a}{2T}$.

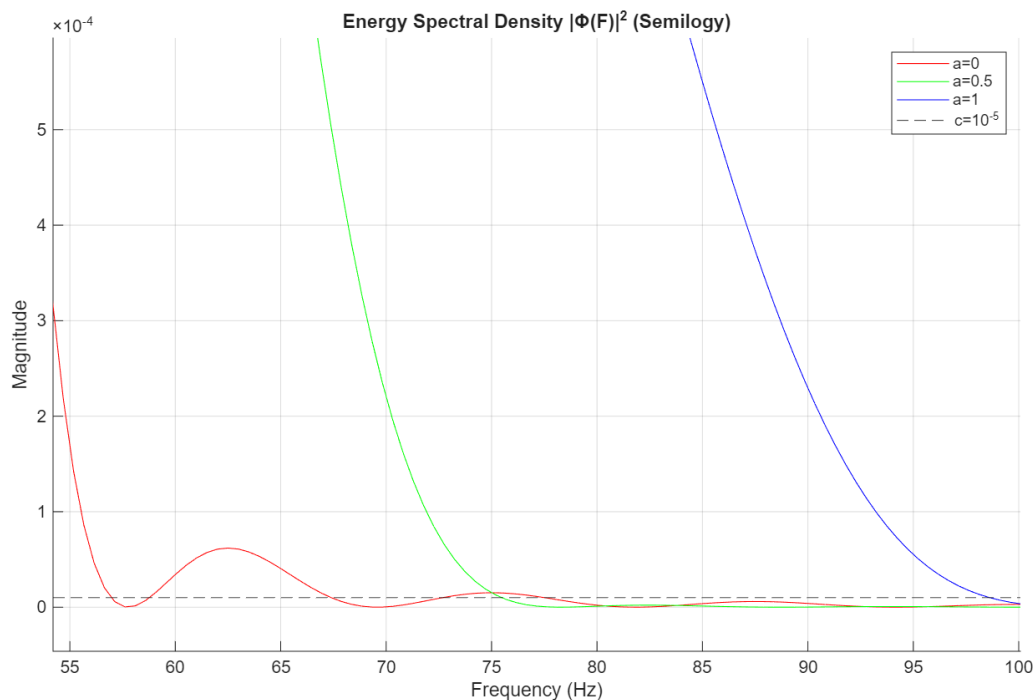
(5) Compute the theoretical bandwidth for each of the three pulses.

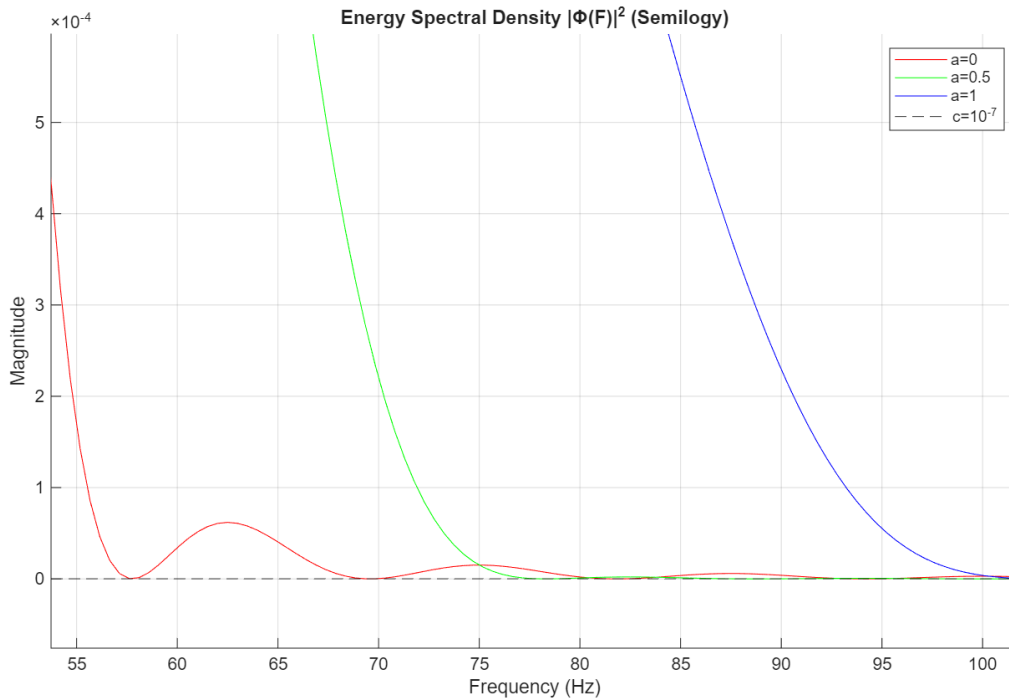
(5) In practice, since truncated pulses have (theoretically) infinite bandwidth, a more **practical** definition of bandwidth is needed. On the common semilogy plot from question A.2, draw a horizontal line with value c (e.g., $c = \frac{T}{10^3}$) and consider values below this line to be “practically zero.” Under this definition, what is the approximate bandwidth of the three pulses (the “Zoom in” feature will be helpful)? Which pulse is more bandwidth-efficient?

How does the bandwidth change if $c = \frac{T}{10^5}$? In this case, which pulse is more efficient?

Theoretical bandwidth ($T = 10^{-2}$):

- $a=0$: $BW = \frac{1+0}{2 \cdot 10^{-2}} = 50 \text{ Hz}$
- $a=0.5$: $BW = \frac{1+0.5}{2 \cdot 10^{-2}} = 75 \text{ Hz}$
- $a=1$: $BW = \frac{1+1}{2 \cdot 10^{-2}} = 100 \text{ Hz}$





Practical BW ($c=10^{-5}$)

- $a=0$: 77.5 Hz
- $a=0.5$: 75.5 Hz
- $a=1$: 98.6 Hz

In this scenario, $a=0.5$ is more bandwidth-efficient than $a=0$, as we expected.

Practical BW ($c=10^{-7}$)

- $a=0$: 116 Hz
- $a=0.5$: 85 Hz
- $a=1$: 102 Hz

As the threshold c becomes smaller, the "practical" bandwidth for all pulses increases because we are now including lower-energy "tails" or side-lobes in the measurement. In this case, the **$a=0.5$** pulse is considered more efficient. While they have a wider "main lobe," their side-lobes decay much faster and stay below the strict threshold.

PART B

B.1 For $T = 10^{-2}$, $A = 4$, $a = 0, 0.5, 1$, and $k = 0, 1, \dots, 2A$ (you will obtain corresponding results for negative k)

1. plot $\phi(t)$ and $\phi(t - kT)$ on the same plot,
2. plot the product $\phi(t)\phi(t - kT)$,
3. numerically approximate the integral of the product $\phi(t)\phi(t - kT)$ using the method described in the notes.

(10) Include in your report the results of steps 1 and 2 for $a = 0.5$ and $k = 0, 1, 2, 3$.

(10) Include in your report the values of the integrals from step 3 for $a = 0, 0.5, 1$ and $k = 0, 1, 2, 3$, and try to explain them.

STEP 1,2

```
%Parametres
```

```
T = 10^-2;
```

```
over = 10;
```

```
A = 4;
```

```
a = 0.5;
```

```
k_values = 0:3;
```

```
fig1 = figure(); hold on; grid on;
```

```
fig2 = figure(); hold on; grid on;
```

```
[phi, t] = srrc_pulse(T, over, A, a);
```

```
for i = 1:length(k_values)
```

```
    k = k_values(i);
```

```
    shift_samples = k * over;
```

```
    phi_shift = [zeros(1, shift_samples), phi(1:end-shift_samples)];
```

```
    product = phi .* phi_shift;
```

```
    figure(fig1);
```

```
    plot(t, phi_shift, 'DisplayName', ['\phi(t-', num2str(k), 'T)']);
```

```
    figure(fig2);
```

```
    plot(t, product, 'DisplayName', ['\phi(t) \phi(t-', num2str(k), 'T)']);
```

```
end
```

```
% Formatting Plot
```

```
figure(fig1);
```

```
xlabel('Time (s)'); ylabel('Amplitude');
```

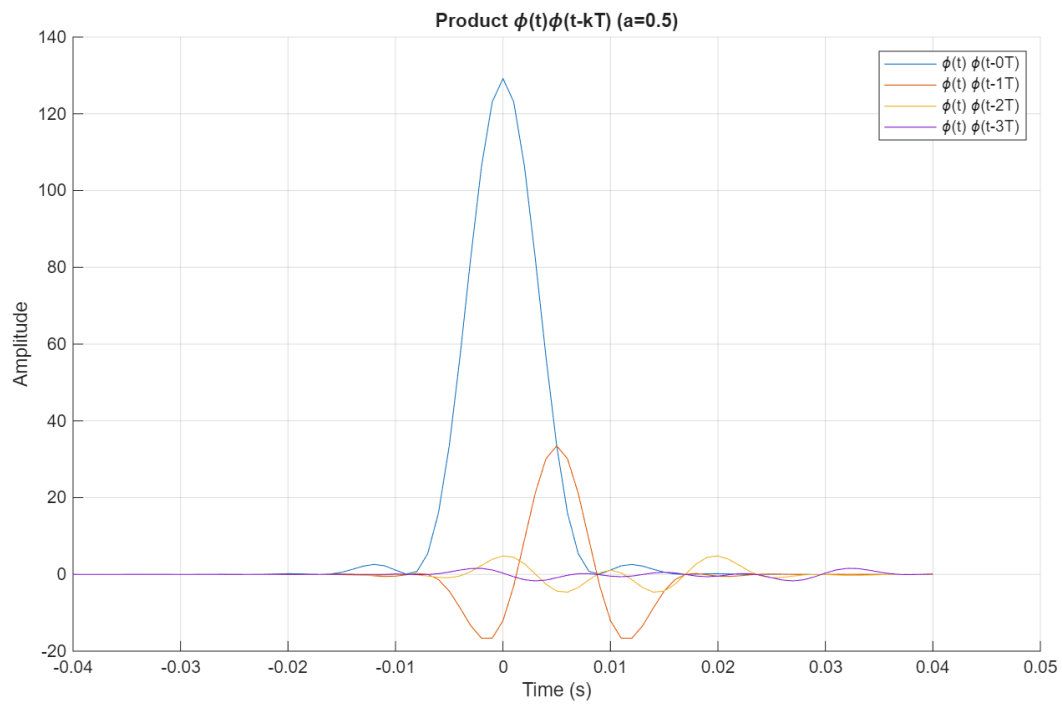
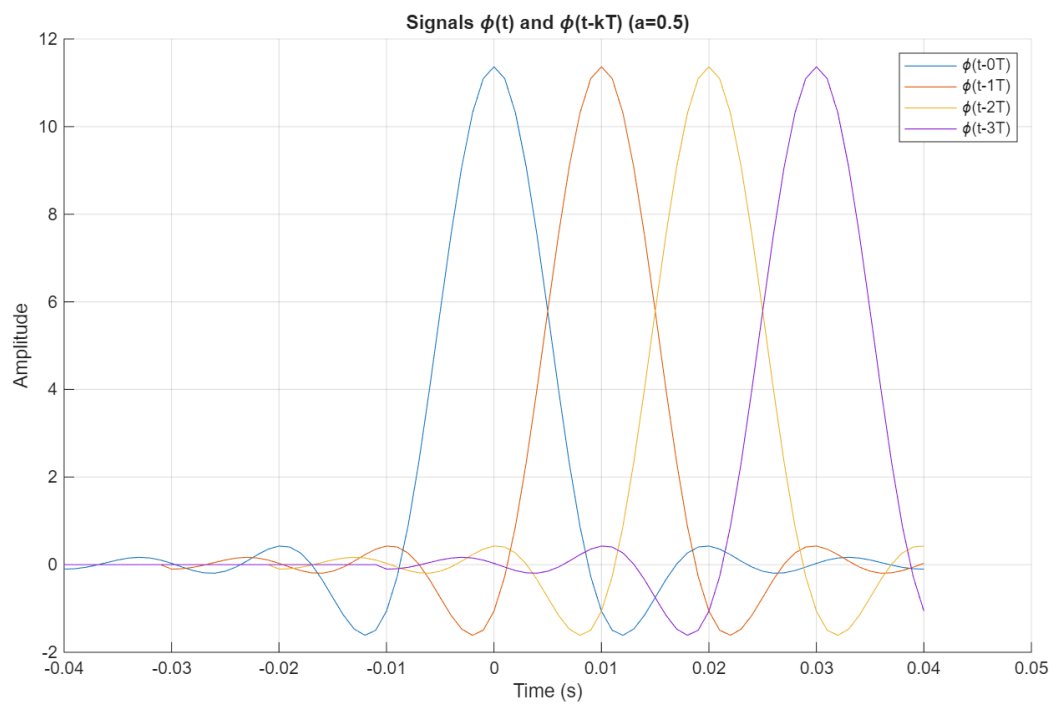
```
title('Signals \phi(t) and \phi(t-kT) (a=0.5)');
```

```
figure(fig2);
```

```
xlabel('Time (s)'); ylabel('Amplitude');
```

```
title('Product \phi(t)\phi(t-kT) (a=0.5)');
```

```
legend('show');
```



STEP 3

% Παράμετροι

T = 10⁻²;

over = 10;

Ts = T/over;

A = 4;

a_values = [0, 0.5, 1];

k_values = 0:3;

for j = 1:length(a_values)

 a = a_values(j);

 [phi, t] = srrc_pulse(T, over, A, a);

 fprintf(['Results for a = %.1f' ...
 '\nResults for a = %.1f \n'], a);

 for i = 1:length(k_values)

 k = k_values(i);

 shift_samples = k * over;

 phi_shift = [zeros(1, shift_samples), phi(1:end-shift_samples)];

 product = phi .* phi_shift;

 integral_val = sum(product) * Ts;

 fprintf('k = %d: Integral = %.4f\n', k, integral_val);

 end

end

Results in command window:

Results for a = 0.0

k = 0: Integral = 0.9747

k = 1: Integral = 0.0290

k = 2: Integral = -0.0349

k = 3: Integral = 0.0461

Results for a = 0.5

k = 0: Integral = 0.9999

k = 1: Integral = 0.0000

k = 2: Integral = 0.0003

k = 3: Integral = -0.0003

Results for a = 1.0

k = 0: Integral = 1.0000

k = 1: Integral = -0.0000

k = 2: Integral = -0.0001

k = 3: Integral = -0.0002

Case $k=0$ (Normality): The integral value are very close to **1** for all values of a . This is because the SRRC pulse is normalized to have unit energy.

Case $k=1, 2, 3$ (Orthogonality): The integral are very close to **0**. The SRRC pulses are designed such that shifts by T are orthogonal to each other.

We notice that as a increases, the values for $k = 1, 2, 3$ get even closer to zero. This is because pulses with higher roll-off factors decay faster in the time domain, meaning the truncation at $A=4$ has a smaller impact on their mathematical properties.

PART C

Finally, we will simulate a baseband PAM system that transmits N bits using 2-PAM modulation. Let $T = 10^{-2}$ sec, $\text{over} = 10$, $a = 0.5$, and $A = 4$ (again, A equals half the length of the truncated pulse measured in symbol periods).

C.1 Generate N bits b_i , for $i = 0, \dots, N - 1$ (e.g., $N = 50, 100$), using the command

$$\mathbf{b} = (\text{sign}(\text{randn}(N, 1)) + 1)/2; \quad (1)$$

C.2 The baseband 2-PAM system is implemented as follows.

1. (10) Write a function

$$\mathbf{X} = \text{bits_to_2PAM}(\mathbf{b});$$

which takes as input the bit sequence $\mathbf{b}$ and outputs the sequence of 2-PAM symbols $\mathbf{X}$ using the mapping:

$$0 \longrightarrow +1,$$

$$1 \longrightarrow -1.$$

2. (10) Simulate the signal

$$X_\delta(t) = \sum_{k=0}^{N-1} X_k \delta(t - kT), \quad (2)$$

using the command

$$\mathbf{X_delta} = 1/T_s * \text{upsample}(\mathbf{X}, \text{over}); \quad (3)$$

Define the appropriate time axis and plot $X_\delta(t)$.

3. (10) Create a *truncated* SRRC pulse $\phi(t)$ (using the parameters above) and simulate the convolution $X(t) = X_\delta(t) \circledast \phi(t)$. Construct the appropriate time axis and plot $X(t)$.¹

4. Assuming an ideal channel, the receiver input is $X(t)$.

(10) Simulate the convolution $Z(t) = X(t) \circledast \phi(-t)$. Plot $Z(t)$ over the corresponding time axis and find the values it takes at time instants kT , for

$k = 0, \dots, N - 1$. Relate these values to X_k , for $k = 0, \dots, N - 1$. Can you explain the phenomenon?

(10) A graphical way to compare $Z(kT)$ with X_k , for $k = 0, \dots, N - 1$, is hold on the plot of $Z(t)$ and execute

```
stem([0 : N - 1] * T, X);
```

where X is the vector of symbols X_k , $k = 0, \dots, N - 1$. What do you observe? Can you explain the phenomenon?

MATLAB code:

```
function X = bits_to_2PAM(b)
    X = zeros(size(b));
    X(b == 0) = 1;
    X(b == 1) = -1;
end

% Parameters
N = 50;
T = 10^-2;
over = 10;
Ts = T/over;
a = 0.5;
A = 4;

%Generate N bits
b = (sign(randn(N,1))+1)/2;
X = bits_to_2PAM(b);

X_delta = zeros(length(X)*over, 1);
X_delta(1:over:end) = X / Ts;

% Time axis for impulse train
t_delta = (0:length(X_delta)-1) * Ts;

figure;
stem(t_delta, X_delta);
title('Impulse Train X_{\delta}(t)');
xlabel('Time (s)'); ylabel('Amplitude');

%Transmit Signal X(t)
[phi, t_phi] = srrc_pulse(T, over, A, a);
X_t = conv(X_delta, phi) * Ts;

% Time axis for X(t)
t_X = (0:length(X_t)-1)*Ts + (t_delta(1) + t_phi(1));
```

```

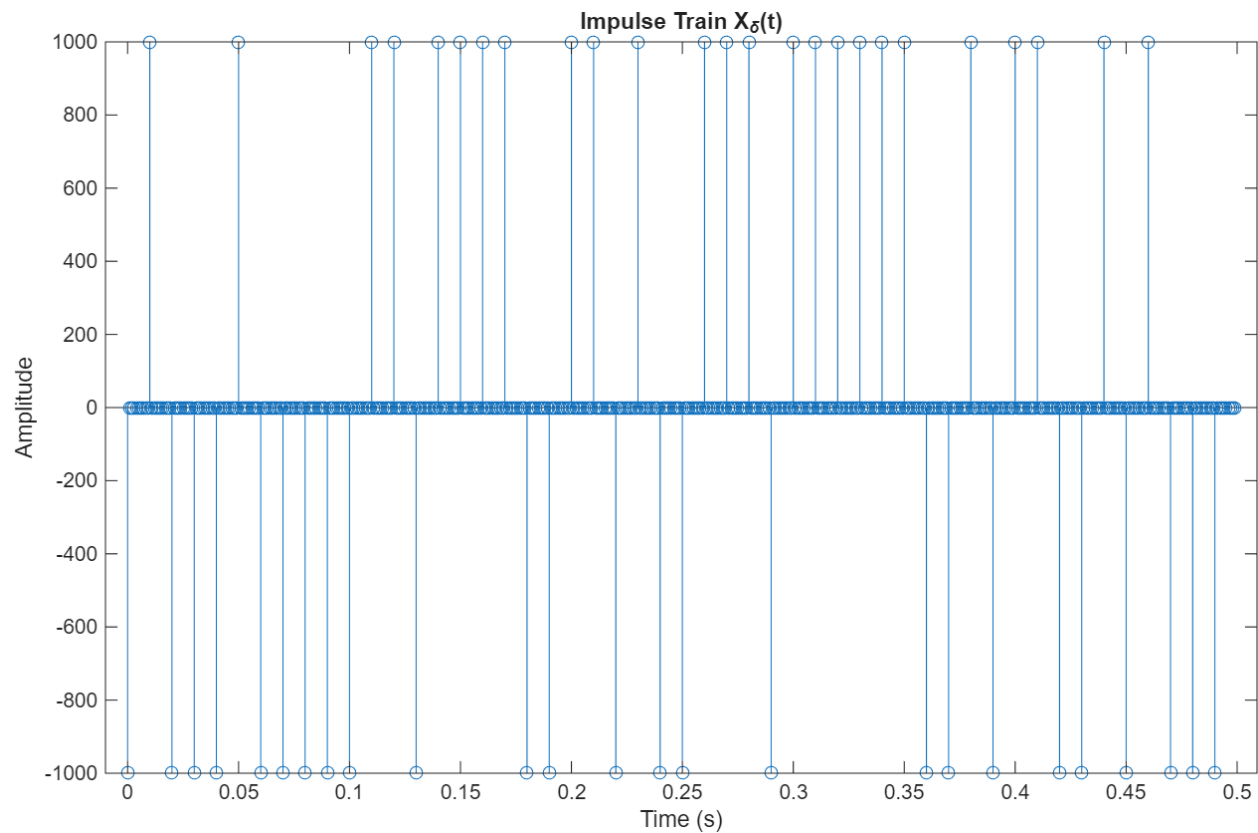
figure;
plot(t_X, X_t);
title('Transmitted Baseband Signal X(t)');
xlabel('Time (s)'); ylabel('Amplitude');
grid on;

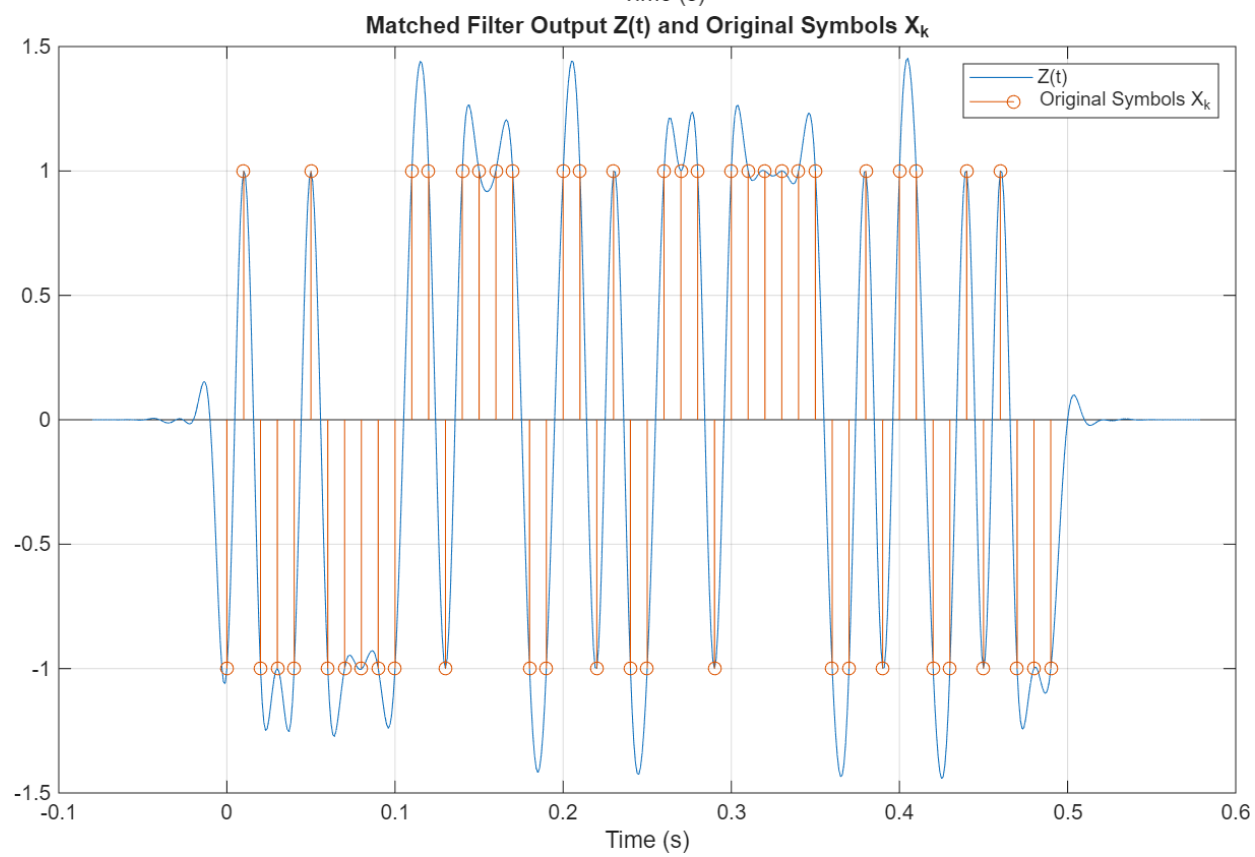
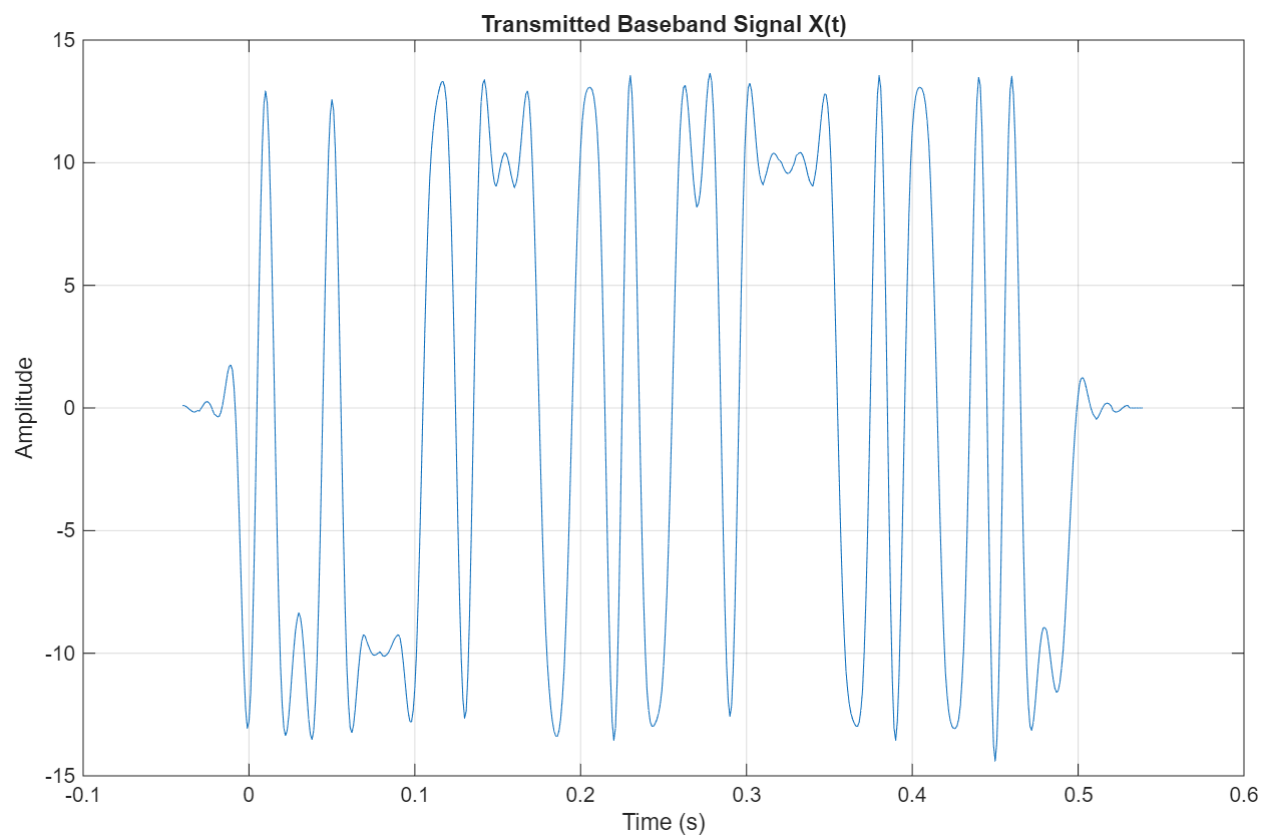
% Receiver Output Z(t)
Z_t = conv(X_t, phi) * Ts; %phi(-t) = phi(t)

% Time axis for Z(t)
t_Z = (0:length(Z_t)-1)*Ts + (t_X(1) + t_phi(1));

figure;
plot(t_Z, Z_t); hold on;
stem((0:N-1)*T, X);
title('Matched Filter Output Z(t) and Original Symbols X_k');
legend('Z(t)', 'Original Symbols X_k');
xlabel('Time (s)'); grid on;

```





The values of the matched-filter output at the sampling instants are approximately equal to the transmitted symbols. This means the receiver recovers the original 2-PAM sequence correctly at the decision times. In an ideal channel, each sample corresponds to one transmitted symbol. This shows that symbol detection works properly at those instants.

This happens because the transmit pulse and the matched filter are designed to work together so that there is no intersymbol interference at the sampling times. Although the pulses overlap in continuous time, their effect from neighboring symbols becomes zero exactly at the decision instants. Therefore, only the desired symbol remains at each sample.

When the transmitted symbols are plotted on top of the matched-filter output, the stem points line up with the waveform at the symbol times. In other words, the sampled values of the output match the transmitted symbol sequence. This provides a clear graphical confirmation that the receiver is recovering the correct data. Any small mismatch is due only to simulation accuracy and truncation effects.